E-WASTE GOVERNANCE MODEL

Collect, pay, securely erase, sort, recycle and educate: design review and implementation route

For the Energy and Environment (UEN008) project at Thapar Institute, Patiala

Group / Roll Nos. / Names: _____

What you proposed, restated

- **Collect** e-waste from consumers and companies and **pay them a modest amount** for each item.
- Run every device through a **data-erasure pipeline**: collection hub, serial/IMEI scan, central wiping station, automated verification, cloud audit report; dead devices go to physical destruction.
- Send the e-waste through **designated, registered recycling stations** where **computer vision** segregates products for recycling.
- Run **awareness campaigns** on reduce and reuse.

Verdict in short

- **The core idea is sound and well aligned with evidence.** India's gap is collection, not recycler capacity; hoarding is driven by data fear and no convenient channel; and a trusted wipe-and-certify step plus payment attacks exactly those two causes.
- **The pipeline as drawn has about a dozen gaps** (section 4): it wipes only after devices travel to the hub, ignores ownership and stolen-phone checks, omits reuse, hazardous items, payment and legal registration, and treats the audit cloud as risk-free although it holds personal data.
- **The weakest link is the payment.** Indian field studies report informal buyers paying at least double what formal recyclers offer, and that certificate markets pay for paperwork more than for collection. "Paying a bit" will not outbid them; it can still work if the draw is trust, convenience, certificates and reuse value, not cash alone.
- **Computer vision is real but belongs at the recycler.** Industrial vision-guided sorting exists (vendor claims of about 80 picks a minute), but a campus hub should use it only for intake classification, as a student demo.
- **For your course project** build a small, safe pilot (not the full national model), measure it, and present the full model as a scalable vision with honest limits (sections 6, 9 and 10).

Reading note. Facts are from the sources listed at the end. Items marked **[inference]** are my reasoning. Legal points need confirmation with your teacher and the Zero Waste Team before you promise anything, and I am not a lawyer. Monetary figures are deliberately symbolic because real prices must come from local quotes.

PART 1: THE GOVERNANCE MODEL

1. Who does what

Devices flow left to right. Money flows right to left: sale and EPR income pay for collection, wiping and payouts. Data flows only as signed certificates.

Legal wrapper: use only registered refurbishers and recyclers; DPDP Act duties for any personal data held; no unregistered buyers.

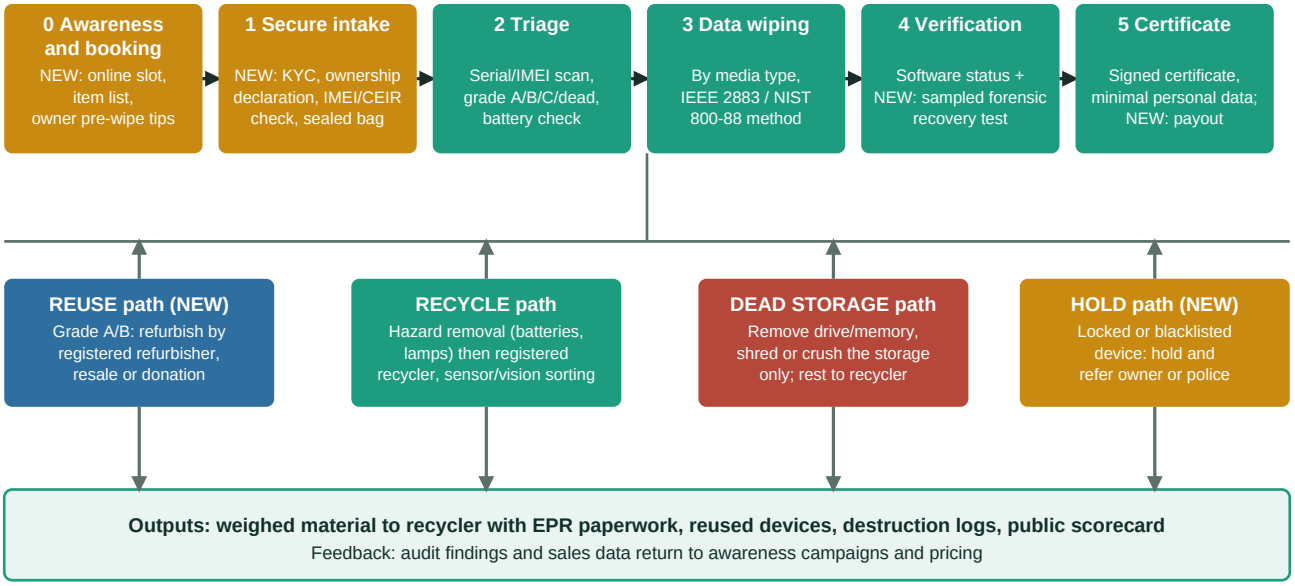
Actor	Role	Key rules
Campus E-Waste Committee	Owns policy: prices, accepted items, safety limits, data-protection rules, partner selection; reviews the audit and publishes results	Includes faculty, Zero Waste Team, IT and legal advice; students hold seats
Student-run hub (operator)	Runs bins, booking, intake, triage, wiping and payouts under faculty supervision	Trained volunteers only; no dismantling; every action logged
Contributors	Students, staff, departments, nearby residents and companies	Sign an ownership and consent declaration; accept terms for data erasure and payment
Registered refurbisher	Repairs and resells or donates reusable devices	Must be registered under the 2022 Rules; gives receipts
Registered recycler	Dismantles and recycles the rest, using vision and sensor sorting at its plant	Must be registered; weighbridge slips, EPR documentation
Independent auditor	Samples wiped devices, checks logs, reconciles weights and payments	Not part of the operator team; reports to the committee

Why a governance model, not just a collection drive

A drive collects once. A governance model fixes who is accountable for each device from drop-off to final material recovery, so that trust (especially on data) can be earned and repeated. The 2022 Rules place duties on producers, refurbishers and recyclers and require bulk consumers such as a university to give e-waste only to registered ones, but they do not require institutions to count what they discard. A campus model supplies that missing accountability on a voluntary basis.

2. Your four-stage pipeline, strengthened

Enhanced pipeline (changes from the original blueprint are marked NEW)



Colour key: amber = new or changed stage; teal = from your blueprint; blue = reuse; red = destruction.

Standards behind the wiping stage

- **NIST SP 800-88 Rev. 2** (published September 2025, replacing Rev. 1) defines three sanitisation levels: **Clear** (logical overwrite or reset), **Purge** (overwrite, block erase or cryptographic erase making recovery infeasible) and **Destroy** (physical). It sets policy; it directs readers to **IEEE 2883-2022** for the technique for each storage type. It expects the outcome to be validated and documented.
- A **factory reset is only Clear**, the lowest level. Simple overwriting is not reliable on SSDs and flash because of wear-levelling and over-provisioned blocks, so purge on flash needs the drive's own secure-erase, sanitise or cryptographic-erase command. Some commentators note independent research found self-encrypting-drive firmware that failed to purge keys fully, so very sensitive data may need destruction as well. Secondary sources also state that degaussing is no longer treated as an approved destroy technique under Rev. 2.

Media	Suggested method (confirm against IEEE 2883)	Verification
Hard disk (HDD)	Overwrite (Clear) is acceptable for low sensitivity; use purge or destroy for sensitive data	Read-back sampling; log certificate
SATA SSD	ATA Secure Erase or cryptographic erase (Purge); plain overwrite is not enough	Tool status plus sample forensic test
NVMe SSD	NVMe Sanitize (block erase or crypto erase) for Purge; "format" only clears the namespace	Controller-reported completion plus sample test
Smartphone / tablet	Remove accounts and activation lock, encrypt, then factory reset (Clear); cryptographic erase through vendor tools where available; destroy if it cannot be reset	Check that device boots to setup; sample forensic check
SD card, USB stick	Treat as flash: purge with the device's method or destroy	Sample check
Network gear, printers, copiers	Reset to defaults, then remove and sanitise any internal storage	Config check
Dead or failed media	Physical destruction of the storage component (shred or crush); keep the rest of the device in the recycling stream	Photo and weight log

PART 2: ROUTE AND IDEAS TO IMPLEMENT FOR YOUR E&E; PROJECT

3. A realistic scope for one semester

The full model needs registrations, funding and equipment that a student group cannot obtain in a semester. The best project is a **small, safe pilot that proves the loop works**, plus a documented vision of the full model.

Layer	Pilot at Thapar (do this)	Full model (describe, do not build)
Collection	Red bins and a booking form; hostel and department champions; one collection day per month	City-wide pickup, company contracts
Payment	Points or certificates first (no cash), with a price-benchmark survey of local scrap dealers and refurbishers	Cash payout through UPI with KYC and fraud controls
Intake and tracking	QR "passport" per item in a spreadsheet, IMEI check on the government portal for phones, ownership declaration	Serial/IMEI scanning at scale, cloud database
Data wiping	Demonstrate on a donated, consented old device using free, standards-aligned tools; hold and escort all other devices to the registered recycler for secure handling	Central wiping station with many parallel ports and audited tools
Reuse	Repair and reuse cafe for chargers, keyboards, simple devices; donation through a registered refurbisher	Refurbishment centre with resale channel
Sorting	Manual sorting by trained volunteers; computer-vision classifier prototype for intake	Vision and sensor sorting at the recycler's plant
Awareness	Orientation talk, QR posters, hostel competition, reels, pledge	Multi-year citywide campaign
Measurement	Baseline audit, weights, survey before and after, energy and carbon estimate	Public dashboard and annual audit

4. Project ideas you can actually build and demonstrate

Demo / deliverable	What to build	Why it helps your marks
E-Waste Passport	Google Form for booking plus a Sheet with auto-generated QR codes and status columns	Shows tracking and chain of custody; easy to run live
Computer-vision classifier	Photograph 150-300 images each of 6-8 classes (phone, laptop, keyboard, charger, battery, circuit board, cable, monitor) and train a small open-source detector; report precision and recall on held-out images	Directly demonstrates the CV step; original data satisfies the "not downloaded" rule
Data-erasure demo	Wipe a donated old hard drive or phone with consent; show before-and-after recovery test and the generated certificate	Shows the trust step end to end
Price calculator	Spreadsheet using the formula in section 5, filled with real local quotes	Shows feasibility and honest economics
Energy and carbon sheet	Measure kWh per wipe with a plug-in power meter; transport kilometres; avoided emissions from reuse (method in section 9)	Ties the project to Energy and Environment
Awareness pack	Poster, 60-second reel, flash cards, pledge card	Evidence of campaign design; mode of presentation

Demo / deliverable	What to build	Why it helps your marks
Pilot results	Kilograms collected, items tagged, items reused, survey change	Real data for the feasibility and applicability segment

5. Pricing and payment: how to pay "a bit" without losing

Offer to contributor = max(S , a x R) - H, with a floor and a cap, where:

- **S** = scrap or material value (weight x the recycler's quoted rate for that category);
- **R** = expected resale or refurbished value after grading; **a** = the share of R the committee passes to the contributor;
- **H** = handling cost per item (wiping time, logistics, testing).

Use real quotes for S and R: ask at least ten local scrap dealers and two refurbishers what they would pay for the same items, and record the numbers. That survey is also your evidence on competitiveness.

Funding source	How it works	Reliability
Resale of refurbished devices	Margin on graded-A/B devices sold or donated at a fee	Medium; needs a registered refurbisher
Recycler payments for material	Recycler pays for weight or material content	Variable with metal prices
EPR-linked income	Partner producers or recyclers pay for collected quantities (certificates)	Uncertain; a field study argues certificates reward paperwork more than collection
Institutional budget or CSR	Campus sustainability budget or corporate CSR funds	Good for a pilot
Voluntary "contribution fee" from companies	Companies pay for certified secure disposal (IT asset disposition)	Strong for B2B because they value the certificate

Controls that prevent abuse

- KYC and bank or UPI payout to the verified owner only; per-person and per-day caps.
- Ownership declaration, and for phones an IMEI check on the Sanchar Saathi / CEIR "Know Your Mobile" service. Blacklisted, duplicate or blocked IMEIs are not purchased and are referred to the owner or police.
- No cash for items of unclear origin; photograph items at intake; keep logs for audit.

> **Competing with the informal market.** One policy study reported that unauthorised recyclers handle most e-waste because they offer at least double the prices of authorised recyclers, and a 2026 field study in Delhi describes cash-paying collectors outbidding formal recyclers because they carry no compliance cost. So a small payment alone will not win. The draw has to be what informal buyers cannot offer: a **data-destruction certificate**, convenience on campus, **fair reuse value** and visible environmental credit. That is a testable claim for your survey.

PART 3: SORTING WITH COMPUTER VISION AND AWARENESS

6. Computer-vision segregation: what it can and cannot do

- **How it works.** Cameras on a conveyor feed a deep-learning model that identifies each item (shape, colour, texture, labels); a robot arm with suction grippers or air jets removes it to the right bin. Sensors such as near-infrared add material information.
- **Evidence.** A leading vendor reports about 80 picks per minute (roughly twice a human sorter), up to about 150 with tandem units, and more than 160 systems installed worldwide. These are company claims. An independent 2026 university proof-of-concept with a camera, microcontrollers and a small

robot arm reached about 81.8% sorting accuracy in controlled tests and noted that dataset diversity, gripping and feedback control need improving.

- **Limits [inference].** Vision sees the outside only, so it cannot tell whether a laptop contains a swollen battery; cluttered, dirty or occluded items reduce accuracy; new product designs need retraining; and installations are expensive and justified only by large volumes.
- **Where it belongs.** At the registered recycler's plant. For your hub, use vision for **intake classification** (a phone-camera app suggests the category and weight class) and keep a person responsible for hazard checks.

7. Awareness campaigns that change behaviour

Evidence from student surveys shows low policy awareness, hoarding and unclear drop-off points, so campaigns should answer three questions: **where do I take it, is my data safe, and what do I get?**

Campaign element	Design	Measure
Launch	Orientation talk and QR posters in every hostel block stating the drop-off point and data promise	Posters seen, QR scans
Competition	Hostel and department leaderboard for kilograms and items reused, with certificates	Items per hostel
Trust message	"See your device wiped" open sessions; show the certificate	Survey: share who trust the process
Repair and reuse	Monthly repair cafe with a faculty mentor	Devices repaired
Reduce first	Short guides: extend phone and laptop life, buy repairable, avoid trade-in churn	Survey on purchase intent
Storytelling	Student reels on where e-waste goes and the health issues in informal recycling	Reach and shares
Feedback	Monthly scorecard posted publicly	Participation trend

Put reduce and reuse ahead of recycling in your messages. Studies of device life cycles show manufacturing dominates a device's footprint (for a laptop 62-70% of life-cycle energy), so keeping devices in use longer saves more than recovering their metals.

PART 4: ADVANTAGES AND CHALLENGES THE SYSTEM SOLVES

Challenge today	How your system helps	Evidence
People hoard old devices	A convenient drop-off with a visible payoff and a trusted wipe	Surveys find storing is the commonest action; data worries and not knowing where to recycle increase it
Fear of data leaks	Certified, standards-based erasure with a per-device certificate	NIST Rev. 2 and IEEE 2883 give a recognised basis
Waste reaches informal recyclers	Registered partners, paperwork and a rewarding formal channel	Reports say most e-waste is handled informally; scrap dealers feed a large share
No visibility of what an institution discards	Passport register and weighed outputs, publicly reported	The 2022 Rules removed bulk-consumer record keeping
Reusable devices are scrapped	Grading and a reuse branch through a registered refurbisher	Reuse avoids most of a device's manufacturing footprint; the Rules recognise refurbishing
Hazard and fire risk from batteries	Battery check at triage and a sealed battery stream	EPA recorded 245 lithium-battery fires at 64 waste facilities (2013-2020)

Challenge today	How your system helps	Evidence
Low awareness	Embedded campaigns with feedback	Only 6.5% of surveyed engineering students knew the policy
Slow manual sorting at recyclers	Vision-guided sorting at scale improves speed and purity	Vendor-reported speeds; limited independent evidence
Businesses need proof of secure disposal	Auditable chain of custody and certificates	Key B2B value; also supports data-protection duties

Additional benefits for Energy and Environment

- Avoided mining and manufacturing energy through reuse and recovery.
- Lower hazard exposure for informal workers if material is diverted to registered recyclers.
- Data for the campus sustainability report and NIRF disclosures.
- A platform for solar-powered operations of the hub, linking to your renewable-energy coursework.

PART 5: FLAWS IN THE ORIGINAL FLOW AND HOW TO FIX THEM

#	Flaw in the blueprint	Why it matters	Fix
1	Wiping happens only after devices reach the hub	Data is exposed during transport and storage	Owner pre-wipe instructions; sealed tamper-evident bags; encrypted devices where possible; logged chain of custody; secure cage at the hub
2	No ownership or legality check	Stolen or blocked phones can be bought; fraud	Ownership declaration, ID, IMEI/CEIR check, refusal of blocked devices
3	Locked devices (activation lock, passcode) not handled	Cannot be wiped or reused	Ask owners to sign out before drop-off; hold locked devices and refer to the owner
4	"If device is dead, destroy" treats the whole device	Destroying everything wastes recoverable material	Remove and destroy only the storage; recycle the rest
5	No reuse branch	Working devices are scrapped	Grade at triage; refurbish through a registered refurbisher
6	Verification only by software	A tool can report success without proof	Add sampled forensic recovery tests by someone independent
7	Cloud audit report may itself hold personal data	Becomes a breach target; Data Protection duties	Store minimal data (device ID, method, time, result); no owner details in the public record; set retention; sign reports
8	No hazard stage	Batteries and lamps cause fires and releases	Remove batteries and lamps at triage into a sealed, labelled stream
9	Payment stage missing	Fraud, dispute and legal exposure	Pricing formula, KYC, caps, receipts, grievance process
10	Registration and legality unspecified	Dealing with unregistered entities is not allowed for registered entities; hub activities may need registration	Partner only with registered refurbishers and recyclers; check registration on the CPCB portal; get legal advice before reselling
11	"Collectors send to designated stations" without traceability	Documented cases of collected waste being sold to unauthorised recyclers	Weighbridge slips, manifests, random audits, GPS or photo logs of handover

#	Flaw in the blueprint	Why it matters	Fix
12	Computer vision placed at the recycler but unmatched to hub	Mismatch of categories and data	Share category codes with the recycler; use vision at intake for classification only
13	Awareness is the last step	Demand and trust are needed first	Start awareness before launch and keep it running with feedback
14	Trade-in incentives may encourage upgrading	Buy-back can shorten device life [inference]	Pair payment with "repair first" messages; pay more for reuse than for scrap

PART 6: LIMITATIONS OF TODAY'S SYSTEM THAT THE MODEL TRIES TO IMPROVE

Limitation	Detail and evidence
Low formal collection	India's formally recycled share is about a third by the CPCB measure and lower against the UN estimate; authorised capacity exceeds generation, so waste is not reaching recyclers
Informal sector dominates	Reported shares from over 50% to 90-95%; informal buyers pay more and carry no compliance cost
EPR rewards paperwork	A 2026 Delhi field study argues certificate markets monetise documentation, not acquisition, so compliance and collection have come apart
Weak enforcement and leakage	Reports of fake certificates and authorised entities selling collected waste to unauthorised recyclers
Institutions have no counting duty	2022 Rules removed bulk-consumer record keeping and annual returns
Data fear blocks disposal	Hoarding rises where people worry about data and lack a trusted channel
Limited reuse	Few structured refurbishing channels for institutions despite refurbishing certificates in the Rules
Battery stream outpaces capacity	Formal lithium-ion recycling is small relative to expected retirements, and mishandled batteries cause fires
Low awareness	Surveyed students show low knowledge of the policy and frequent dumping or storing
Poor data	National figures are modelled and differ (1.6 Mt CPCB vs 4.1 Mt UN); state data are not published
Poor Indian recovery rates (older study)	A Bengaluru study reported gold recovery of 8-18% in Indian recyclers against over 95% internationally; check for newer data before quoting

PART 7: DISADVANTAGES AND RISKS OF THE PROPOSED SYSTEM

An honest review that you can use in the "feasibility and applicability" segment and in the viva:

Disadvantage	Explanation	Mitigation
Cannot out-pay the informal market	Informal buyers can pay more; a small payment may not attract high-value items, leaving the formal hub the low-value, costly waste	Compete on certificates, convenience and reuse; seek DRS-style funding; prioritise institutions and companies that value proof
Data-breach liability	Holding devices before wiping makes the hub responsible. Under India's DPDP Act and Rules (notified November 2025; main duties enforceable from May 2027) failure to keep reasonable security can attract penalties up to Rs 250 crore, and failure to notify a breach up to Rs 200 crore	Secure cage, access logs, encryption, sealed bags, breach-response plan; get legal advice on whether the hub is a data fiduciary or processor

Disadvantage	Explanation	Mitigation
Wiping is slow and imperfect	Large drives take hours; SSD and phone wiping depends on vendor support; locked devices cannot be wiped	Parallel stations, purge methods per IEEE 2883, destroy storage when in doubt
False sense of security	A certificate proves a process ran, not that no data remains	Sampled forensic checks and independent audit
Fraud and adverse selection	Payment can attract stolen, counterfeit or imported devices and dumping of worthless items	KYC, IMEI check, caps, photos, refuse unclear items
Cost and funding gap	Labour, tools, storage, insurance, audit and payouts need money; EPR income is uncertain	Pilot on budget; recycler partnership; resale margin; CSR
Legal and registration burden	Rules 2022, Battery Rules, DPDP, GST and local approvals; a student hub is not a registered entity	Operate only as a collection point feeding registered partners; get written approvals
Computer vision cost and accuracy	Expensive, needs data and maintenance; accuracy falls on clutter; cannot see inside devices	Use at recycler scale only; use vision for intake demo
Trade-in rebound	Paying for old devices can encourage faster replacement	Emphasise repair and longer use; price reuse above scrap
Livelihood impacts	Formalisation can displace informal collectors	Include them as aggregators or workers, as Thapar did with its waste staff
Single point of failure	Centralised hub and cloud tools can fail or be attacked	Backups, minimal data, manual fallback
Environmental costs	Transport, energy for wiping and destroyed storage	Consolidate trips, measure kWh, destroy only storage
Small-scale pilot may not generalise	A campus is not a city; student volunteers rotate	Document results as a pilot, hand over to a standing club
Safety	Battery fires, sharps, dust, lifting	Training, PPE, fire-safe battery storage, no dismantling

Bottom line for your presentation. The model addresses real causes (no trusted channel, data fear, no reuse, no visibility), but its economics and legal exposure are the hardest parts. The honest recommendation is: pilot on campus without cash, prove collection and trust, measure, then add payment and company contracts with legal approval.

PART 8: ENERGY AND ENVIRONMENT CALCULATIONS FOR YOUR PROJECT

9. How to quantify the benefit

- **Avoided emissions from reuse** = number of devices reused x emissions of manufacturing a comparable new device x displacement factor (the fraction of reused devices that really replace a new purchase). Published figures for scale: manufacturing a laptop emits about 227-270 kg CO2 (ASU study); an industry body estimates manufacturing and materials account for 70-90% of a smartphone's life-cycle emissions. Use a conservative displacement factor (for example 0.5) and state it.
- **Hub energy** = measured kWh per wipe x devices wiped x grid emission factor. Measure with a plug-in power meter; this is small and should be reported honestly.
- **Transport emissions** = kilometres x vehicle emission factor; consolidate trips.
- **Material recovery** = kilograms handed to registered recyclers x recovery fractions from the recycler's report; do not claim more than the recycler documents.

- **Net benefit** = avoided emissions - hub energy - transport emissions; present it with the assumptions and ranges.

10. Plan for the semester

Weeks	Activities	Output
1-2	Confirm topic, permissions, safety rules; read the extended report; draft the model	Approved scope
3-4	Audit and survey; local price-benchmark survey; select partners (registered recycler, refurbisher)	Baseline data and quotes
5-7	Install bins; booking form and QR register; hostel champions; awareness launch	Pilot live
8-10	Intake and triage; repair cafe; wipe demo; CV classifier training and testing	Demo-ready prototypes
11-12	Second survey; weigh outputs; energy and carbon sheet; reconcile with recycler receipts	Results
13	Record the video; prepare slides; rehearse viva	Submission

Mapping to the 10-minute video

Segment	Time	Content from this plan	Presenters (7-8 members)
Problem statement and concern	1.5 min	Scale, hoarding, Thapar gap, data fear	1-2 members
Existing solutions	1.5 min	Rules 2022, EPR, informal sector, campus drives; limitations (Part 6)	1 member
Idea	5 min	Governance model; pipeline; pricing; wipe demo; CV demo; awareness	3-4 members
Feasibility and applicability	2 min	Pilot results, limits and disadvantages (Part 7), scaling	1-2 members

Remember the course rules: real-time footage, each member presents a part, group and roll numbers at the start, no animation, no downloaded material, under 25 MB.

PART 9: VIVA QUESTIONS AND SOURCES

Question	Short answer
Why pay people for e-waste?	Because informal buyers pay, and without a payoff or trust, people hoard or sell informally. Our payment is paired with certified data erasure, convenience and reuse value.
Can you out-pay kabadiwalas?	Not on price alone. Studies report informal buyers pay at least double. We compete on data certificates, convenience and reuse value, and we test this with a price survey.
Is a factory reset enough?	No. NIST treats it as the lowest level (Clear). For flash storage we use the drive's secure-erase or cryptographic-erase commands, and destroy storage that cannot be sanitised.
What if the device is stolen?	We check the IMEI on the government portal and refuse blocked or blacklisted devices; owners sign a declaration.
Is computer vision necessary?	Not at campus scale. It is valuable at the recycler's plant; we use it only to classify items at intake and as a demonstration.

Question	Short answer
What are your legal risks?	Handling personal data (DPDP Act), dealing only with registered refurbishers and recyclers (2022 Rules), and battery safety. We operate as a collection point feeding registered partners.
What is the main weakness?	Economics: payments, wiping labour and audit cost money and EPR income is uncertain. We start with a no-cash pilot.
How does it help the environment?	Reuse avoids new manufacturing, registered recycling avoids informal pollution and mining, and awareness reduces waste; we quantify with a transparent method and report assumptions.

Sources

- NIST SP 800-88 Rev. 2 (September 2025) and IEEE 2883-2022, as summarised at phonecheck.com, ziperase.com, bitraser.com, killdisk.com, technologymatch.com, inventivehq.com (vendor and commentary sources; read the standards themselves before quoting).
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- Computer-vision sorting: IEEE Spectrum article by AMP Robotics' founder; recyclinginside.com; trellis.net; Prakash et al. (2026) in *Waste* (USP repository); arXiv 2404.01116.
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- E-Waste (Management) Rules 2022: mondaq.com (Khaitan; AMS Shardul), lakshmisri.com, lkslaw.com, pmfias.com, taxguru.in. Battery Waste Management Rules 2022: indiaenvironmentportal.org.in.
- US EPA (2021) analysis of lithium-ion battery fires in waste management and recycling; pinfa.eu summary.
- Life-cycle evidence: ASU laptop study (Williams) via Network World; GSMA via Guardian Nigeria; E3S Web of Conferences 349 (2022).
- Thapar and campus context: Education Post (27 Aug 2024); NIRF 2025 SDG submission; sources listed in the extended research report.

Prepared 12 October 2026 from publicly available sources. Re-check every figure and legal point against the original source before quoting it.